

Given a **joint probability** $p(x, y)$, where $x \in \{0,1\}$ and $y \in \{0,1\}$:

$$p(x = 0, y = 0) = a$$

$$p(x = 0, y = 1) = b$$

$$p(x = 1, y = 0) = c$$

$$p(x = 1, y = 1) = d$$



For $p(x, y)$ to be a valid probability:

$$a + b + c + d = 1$$

$p(x, y)$	$x=0$	$x=1$
$y=0$	a	c
$y=1$	b	d

Find the **marginal probability** $p(x)$:

$$p(x) = \sum_y p(x, y) = p(x, y = 0) + p(x, y = 1), \text{ i.e.}$$

$$p(x = 0) = p(x = 0, y = 0) + p(x = 0, y = 1) = a + b,$$

$$p(x = 1) = p(x = 1, y = 0) + p(x = 1, y = 1) = c + d$$

Check that $p(x)$ is a valid probability: $(a + b) + (c + d) = 1$

$p(x)$	$x=0$	$x=1$
	$a + b$	$c + d$

Find the **conditional probability** $p(y | x)$:

$$p(y | x) = \frac{p(x, y)}{p(x)}$$

$$p(y = 0 | x = 0) = \frac{p(x = 0, y = 0)}{p(x = 0)} = \frac{a}{a + b}$$

$$p(y = 1 | x = 0) = \frac{p(x = 0, y = 1)}{p(x = 0)} = \frac{b}{a + b}$$

For $p(y | x = 0)$ to be a valid probability: $\frac{a}{a+b} + \frac{b}{a+b} = 1$

$$p(y = 0 | x = 1) = \frac{p(x = 1, y = 0)}{p(x = 1)} = \frac{c}{c + d}$$

$$p(y = 1 | x = 1) = \frac{p(x = 1, y = 1)}{p(x = 1)} = \frac{d}{c + d}$$

For $p(y | x = 1)$ to be a valid probability: $\frac{c}{c+d} + \frac{d}{c+d} = 1$

$p(y x)$	$x=0$	$x=1$
$y=0$	$\frac{a}{a+b}$	$\frac{c}{c+d}$
$y=1$	$\frac{b}{a+b}$	$\frac{d}{c+d}$

Find the $p(x | y)$ using **Bayes' rule**:

$$p(x | y) = \frac{p(y | x)p(x)}{\sum_x p(y | x)p(x)}$$

$$p(x = 0 | y = 0) = \frac{p(y = 0 | x = 0)p(x = 0)}{p(y = 0 | x = 0)p(x = 0) + p(y = 0 | x = 1)p(x = 1)}$$

$$p(x = 1 | y = 0) = \frac{p(y = 0 | x = 1)p(x = 1)}{p(y = 0 | x = 0)p(x = 0) + p(y = 0 | x = 1)p(x = 1)}$$

$$p(x = 0 | y = 1) = \frac{p(y = 1 | x = 0)p(x = 0)}{p(y = 1 | x = 0)p(x = 0) + p(y = 1 | x = 1)p(x = 1)}$$

$$p(x = 1 | y = 1) = \frac{p(y = 1 | x = 1)p(x = 1)}{p(y = 1 | x = 0)p(x = 0) + p(y = 1 | x = 1)p(x = 1)}$$

$$p(x = 0 \mid y = 0) = \frac{p(y = 0 \mid x = 0)p(x = 0)}{p(y = 0 \mid x = 0)p(x = 0) + p(y = 0 \mid x = 1)p(x = 1)}$$

$$= \frac{\left(\frac{a}{a+b}\right)(a+b)}{\left(\frac{a}{a+b}\right)(a+b) + \left(\frac{c}{c+d}\right)(c+d)} = \frac{a}{a+c}$$

$$p(x = 1 \mid y = 0) = \frac{p(y = 0 \mid x = 1)p(x = 1)}{p(y = 0 \mid x = 0)p(x = 0) + p(y = 0 \mid x = 1)p(x = 1)}$$

$$= \frac{\left(\frac{c}{c+d}\right)(c+d)}{\left(\frac{a}{a+b}\right)(a+b) + \left(\frac{c}{c+d}\right)(c+d)} = \frac{c}{a+c}$$

$$p(x = 0 \mid y = 1) = \frac{p(y = 1 \mid x = 0)p(x = 0)}{p(y = 1 \mid x = 0)p(x = 0) + p(y = 1 \mid x = 1)p(x = 1)}$$

$$= \frac{\left(\frac{b}{a+b}\right)(a+b)}{\left(\frac{b}{a+b}\right)(a+b) + \left(\frac{d}{c+d}\right)(c+d)} = \frac{b}{b+d}$$

$$p(x = 1 \mid y = 1) = \frac{p(y = 1 \mid x = 1)p(x = 1)}{p(y = 1 \mid x = 0)p(x = 0) + p(y = 1 \mid x = 1)p(x = 1)}$$

$$= \frac{\left(\frac{d}{c+d}\right)(c+d)}{\left(\frac{b}{a+b}\right)(a+b) + \left(\frac{d}{c+d}\right)(c+d)} = \frac{d}{b+d}$$

We have $p(x | y) = \frac{p(y|x)p(x)}{\sum_x p(y|x)p(x)}$ in the following table:

$p(x y)$	$x=0$	$x=1$
$y=0$	$\frac{a}{a+c}$	$\frac{c}{a+c}$
$y=1$	$\frac{b}{b+d}$	$\frac{d}{b+d}$

Check that $\sum_x p(x | y = 0) = 1$:

$$\begin{aligned}\sum_x p(x | y = 0) &= p(x = 0 | y = 0) + p(x = 1 | y = 0) \\ &= \frac{a}{a+c} + \frac{c}{a+c} = 1. \quad \square\end{aligned}$$

Check that $\sum_x p(x | y = 1) = 1$:

$$\begin{aligned}\sum_x p(x | y = 1) &= p(x = 0 | y = 1) + p(x = 1 | y = 1) \\ &= \frac{b}{b+d} + \frac{d}{b+d} = 1. \quad \square\end{aligned}$$